

# VIT GradeTrack

## Project Report

A simple command-line Python project for managing subjects, marks and attendance

|                            |                                           |
|----------------------------|-------------------------------------------|
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| <b>Programme</b>           | B.Tech CSE (AI & ML)                      |
| <b>Institute</b>           | VIT Bhopal University                     |
| <b>Course</b>              | CSE1021 – Problem Solving and Programming |
| <b>Project</b>             | VIT GradeTrack                            |
| <b>Project Type</b>        | Python Command-Line Application           |

### GitHub Repository

<https://github.com/tusharr-012/VIT-GradeTrack>

## 1. Introduction

VIT GradeTrack is a small Python project made to keep basic academic information in one place. The idea is simple: a student can add subjects, enter marks, check attendance and see a few basic calculations without using a complicated application.

The project runs completely from the terminal. A numbered menu is shown to the user and the selected operation is performed. The main purpose of making this project was also to use the Python concepts learned in the Problem Solving and Programming course in one working program.

## 2. Problem Statement

During a semester, a student has to keep track of different subjects, CAT marks, TEE marks and attendance. If these details are handled separately, it can become difficult to find or calculate the required information quickly.

The project solves this problem at a basic level by providing a single command-line program where subject details can be entered and viewed whenever the program is running.

## 3. Objectives

- To make a simple Python program that can be run from the command line.
- To store subject names and subject codes using basic Python data structures.
- To enter CAT1, CAT2 and TEE marks for subjects.
- To check attendance percentage and show the attendance status.
- To calculate total marks and basic marks statistics.
- To provide search and delete options for subjects.
- To practice functions, modules, lists, dictionaries, loops, conditions and exception handling.

## 4. Main Features

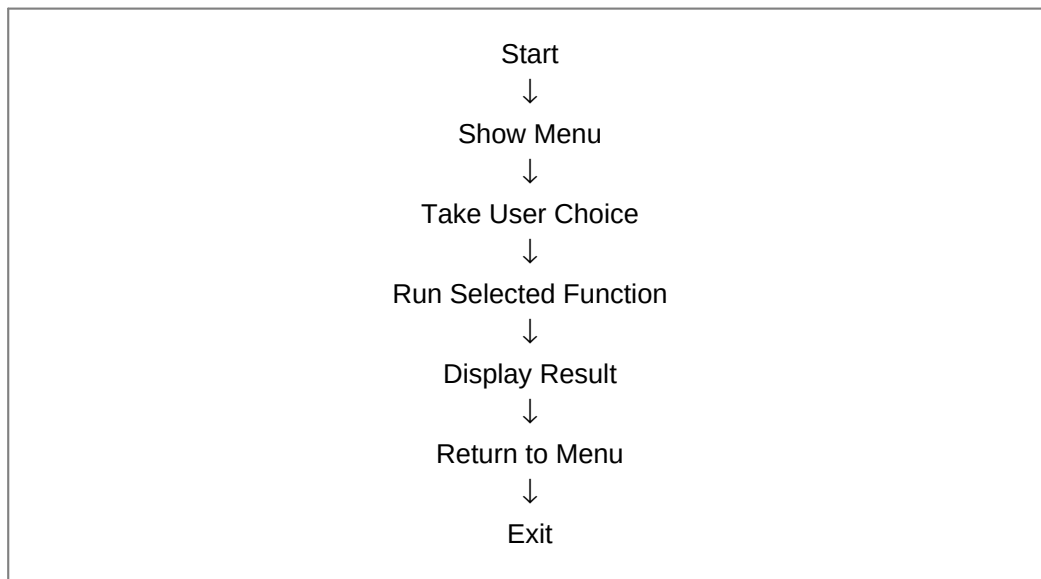
| Feature                 | What it does                                              |
|-------------------------|-----------------------------------------------------------|
| <b>Add Subject</b>      | Adds a subject name and subject code to the subject list. |
| <b>View Subjects</b>    | Displays the subjects that have been added.               |
| <b>Enter Marks</b>      | Takes CAT1, CAT2 and TEE marks for a selected subject.    |
| <b>Attendance Check</b> | Takes attendance percentage and checks it against 75%.    |
| <b>Marks Analysis</b>   | Shows total, highest, lowest and average marks.           |
| <b>Search Subject</b>   | Finds a subject using its subject code.                   |
| <b>Delete Subject</b>   | Removes a subject from the current list.                  |
| <b>Input Validation</b> | Checks marks and handles invalid numerical input.         |

## 5. How the Program Works

When the program starts, the main menu is displayed. The user enters a number from 1 to 8. The corresponding function is then called. After the operation is completed, the menu appears again. This

continues until the user selects Exit.

## Basic Workflow



## 6. Project Structure

The project is divided into small Python files instead of keeping everything in one file. Each file has a simple purpose.

| File                            | Purpose                                                      |
|---------------------------------|--------------------------------------------------------------|
| <b>main.py</b>                  | Displays the menu and controls the program flow.             |
| <b>subject.py</b>               | Handles subjects, marks, attendance, searching and deleting. |
| <b>calculator.py</b>            | Calculates total marks and performs marks analysis.          |
| <b>validator.py</b>             | Checks whether entered marks are within the allowed range.   |
| <b>tests/test_calculator.py</b> | Contains basic tests for the marks calculation.              |
| <b>README.md</b>                | Explains the project and how to run it.                      |
| <b>statement.md</b>             | Contains the project statement and main functions.           |
| <b>docs/design.md</b>           | Contains the project design and workflow details.            |

## 7. Data Structures Used

The project mainly uses simple Python data structures so that the code stays easy to understand.

**List:** The main subjects list stores all subjects added by the user.

**Dictionary:** Each subject is stored as a dictionary containing details such as name, code and marks.

**String:** Subject names, subject codes and menu input are handled as strings.

**Integer:** Marks are stored as integers.

**Float:** Attendance percentage is taken as a float so values such as 74.5 can be entered.

## 8. Python Concepts Used

- Variables and basic data types
- Input and output
- Strings
- Lists
- Dictionaries
- If, elif and else statements
- For and while loops
- Functions
- Function parameters and return values
- Modules and imports
- Exception handling
- Basic searching and calculations

## 9. Functional Requirements

- The user can add a subject with its name and subject code.
- The user can view all added subjects.
- The user can enter CAT1, CAT2 and TEE marks.
- The user can check attendance percentage.
- The program can calculate total marks.
- The program can show highest, lowest and average total marks.
- The user can search for a subject using its code.
- The user can delete a subject.
- The program gives a message for an invalid menu choice.
- The program handles invalid numerical input for marks.

## 10. Non-Functional Requirements

**Usability:** The menu should be simple enough for a user to understand.

**Reliability:** Wrong marks or menu input should not make the program stop unexpectedly.

**Maintainability:** The code is split into separate files and functions.

**Performance:** The program should respond quickly for a normal number of subjects.

**Readability:** The code uses simple function and variable names.

## 11. Marks and Attendance

The program takes CAT1 marks out of 50, CAT2 marks out of 50 and TEE marks out of 100. For the Marks Analysis option, these values are added to get a total out of 200. The program also calculates the highest, lowest and average total for subjects that have marks entered.

For attendance, the program takes a percentage from 0 to 100. If the value is 75% or above, it displays that the attendance requirement is met and shows 5/5 attendance marks. If the value is below 75%, it informs the user that full attendance marks are not applicable.

The project does not calculate an official VIT grade or use an invented grade conversion formula.

## 12. Input Validation and Error Handling

Marks are checked before they are stored. CAT1 and CAT2 must be between 0 and 50, while TEE must be between 0 and 100. If the user enters text instead of a number, the program shows an error message and asks for the input again.

Attendance is also checked so that values below 0 or above 100 are not accepted. If a subject code is not found, the program displays a simple 'Subject not found' message.

## 13. Testing

Basic testing was done using pytest. The current test file contains two tests for the total marks calculation function.

| Test           | Expected Result            |
|----------------|----------------------------|
| Normal marks   | Correct total is returned. |
| All marks zero | Total returned is 0.       |

Test command: `python -m pytest`

Result during final project testing: **2 tests passed**.

The main Python files were also checked with Python's `compileall` command and no syntax errors were reported.

## 14. Limitations

- The data is stored only while the program is running.
- All entered data is lost when the program is closed.
- The program works through the terminal and does not have a graphical interface.
- There is no database or online storage.
- The project does not calculate an official VIT grade.

## 15. Conclusion

VIT GradeTrack is a basic academic tracking program made using Python. It combines the concepts learned in the course into one working project. The program can manage subjects, take marks, check attendance and perform simple calculations.

The project is intentionally kept simple so that the code and its working can be understood easily. It also gives a practical example of how functions, modules, lists, dictionaries, conditions, loops and input validation can be used together in a small application.

## 16. GitHub Repository

The complete project is available in the following public GitHub repository:

## 17. Final Project Details

|                              |                                        |
|------------------------------|----------------------------------------|
| <b>Programming Language</b>  | Python                                 |
| <b>Application Type</b>      | Command-line application               |
| <b>Data Storage</b>          | In-memory while the program is running |
| <b>Testing Tool</b>          | pytest                                 |
| <b>Source Control</b>        | Git / GitHub                           |
| <b>Repository Visibility</b> | Public                                 |

### Output 1 – Add Subject

```
PS C:\Users\tdhin\Documents\VIT_Gradetrack Projects> python main.py

=====
VIT GRADETRACK
=====
1. Add Subject
2. View Subjects
3. Enter Marks
4. Attendance Check
5. Marks Analysis
6. Search Subject
7. Delete Subject
8. Exit
Enter your choice: 1
Enter subject name: Problem Solving and Programming
Enter subject code: CSE1021
Subject added successfully!
```

### Output 2 – View Subjects

```
=====
VIT GRADETRACK
=====
1. Add Subject
2. View Subjects
3. Enter Marks
4. Attendance Check
5. Marks Analysis
6. Search Subject
7. Delete Subject
8. Exit
Enter your choice: 2

----- SUBJECTS -----
Subject: Problem Solving and Programming
Code: CSE1021
Marks: Not entered yet
```

### Output 3 – Enter Marks

```
=====
VIT GRADETRACK
=====
1. Add Subject
2. View Subjects
3. Enter Marks
4. Attendance Check
5. Marks Analysis
6. Search Subject
7. Delete Subject
8. Exit
Enter your choice: 3
Enter subject code: CSE1021
Enter CAT1 marks (out of 50): 40
Enter CAT2 marks (out of 50): 42
Enter TEE marks (out of 100): 75
Marks added successfully!
```

### Output 4 – Attendance Check

```
=====
VIT GRADETRACK
=====
1. Add Subject
2. View Subjects
3. Enter Marks
4. Attendance Check
5. Marks Analysis
6. Search Subject
7. Delete Subject
8. Exit
Enter your choice: 4
Enter subject code: CSE1021
Enter attendance percentage: 92
Attendance: 92.0 %
Attendance requirement met.
Attendance marks: 5 / 5
```

### Output 5 – Marks Analysis

```
=====
VIT GRADETRACK
=====
1. Add Subject
2. View Subjects
3. Enter Marks
4. Attendance Check
5. Marks Analysis
6. Search Subject
7. Delete Subject
8. Exit
Enter your choice: 5

----- MARKS ANALYSIS -----
Subject: Problem Solving and Programming
Code: CSE1021
Total: 157 / 200

Highest Total: 157 / 200
Lowest Total: 157 / 200
Average Total: 157.0 / 200
```

### Output 6 – Search Subject

```
=====
VIT GRADETRACK
=====
1. Add Subject
2. View Subjects
3. Enter Marks
4. Attendance Check
5. Marks Analysis
6. Search Subject
7. Delete Subject
8. Exit
Enter your choice: 6
Enter subject code to search: CSE1021

Subject found!
Subject: Problem Solving and Programming
Code: CSE1021
CAT1: 40 / 50
CAT2: 42 / 50
TEE: 75 / 100
```

### Output 7 – Delete Subject

```
=====
VIT GRADETRACK
=====
1. Add Subject
2. View Subjects
3. Enter Marks
4. Attendance Check
5. Marks Analysis
6. Search Subject
7. Delete Subject
8. Exit
Enter your choice: 7
Enter subject code to delete: CSE1021
Subject deleted successfully!
```

### Output 8 – Exit

```
=====
VIT GRADETRACK
=====
1. Add Subject
2. View Subjects
3. Enter Marks
4. Attendance Check
5. Marks Analysis
6. Search Subject
7. Delete Subject
8. Exit
Enter your choice: 8
Thank you for using VIT GradeTrack!
```